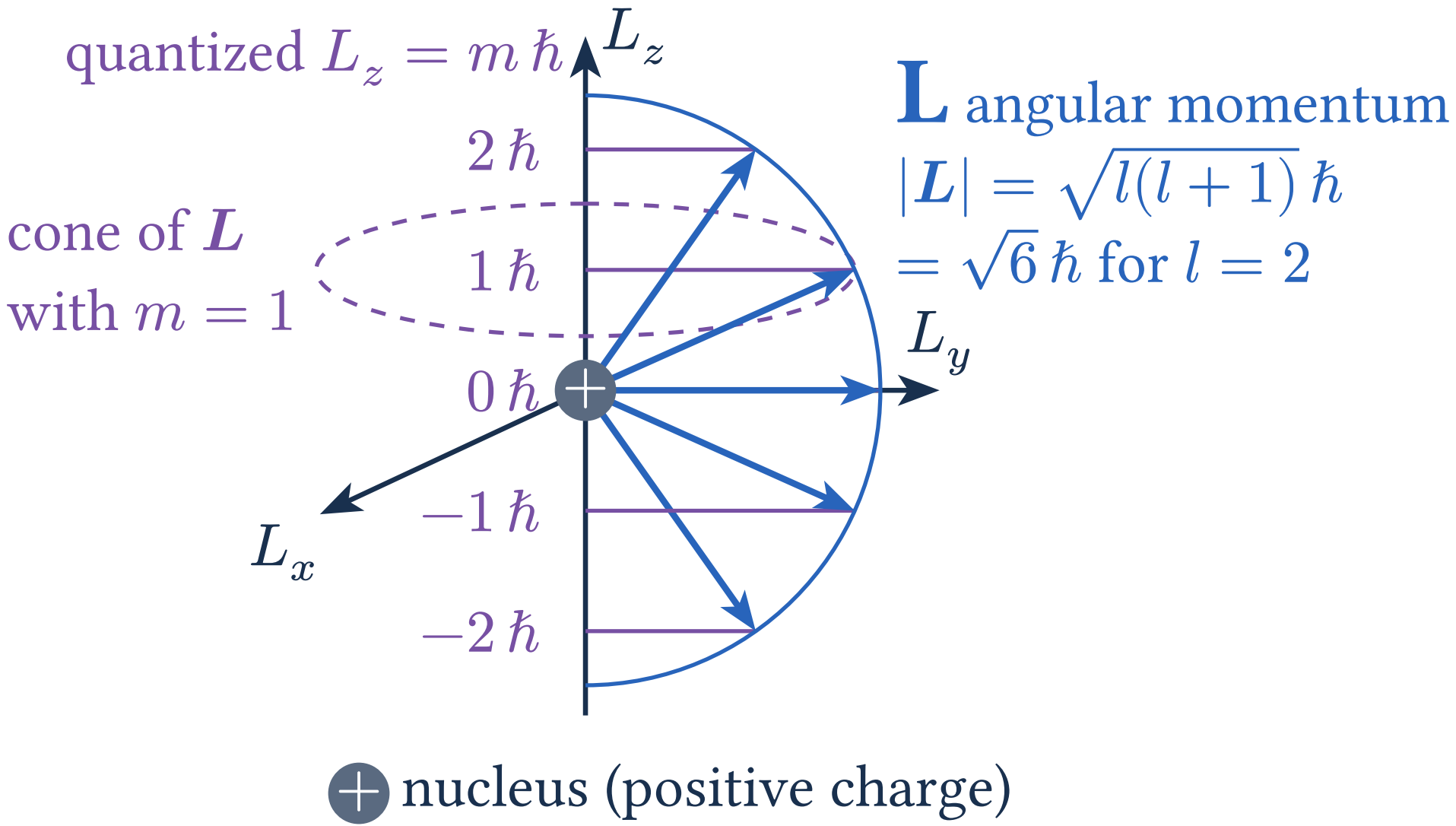
quantized $L_z = m\,\hbar$
L_z
$2\,\hbar$
$1\,\hbar$
$0\,\hbar$
$-1\,\hbar$
$-2\,\hbar$
cone of $\boldsymbol{L}$
with $m = 1$
L_y
L_x
$\mathbf{L}$ angular momentum
$|\boldsymbol{L}| = \sqrt{l(l+1)}\,\hbar$
$= \sqrt{6}\,\hbar$ for $l = 2$
nucleus (positive charge)